

1 (i) $x^2 + 2x + 1 = (x+1)^2$

$\Rightarrow B = 1$

by comparison or long division

$C = 3$

$A = 4$

(ii) *translate left*
1st Shift the graph 1 unit in the negative x-direction; we have the graph of

$y = \frac{1}{(x+1)^2}$

stretch
2nd ~~Enlargement~~ the graph by factor 4 in the vertical direction. We have

the graph of $y = \frac{4}{(x+1)^2}$

3rd shift/(translate) the graph 3 units in the positive y direction we

$\frac{4}{(x+1)^2} + 3$

2 Let the equation of the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ since the circle passes through $(-1, 0)$, $(1, 2)$ and $(-5, 4)$

It must satisfy the equations

$1 + 0 - 2g + c = 0$

$1 + 4 + 2g + 4f + c = 0$

$25 + 16 - 10g + 8f + c = 0$

(i)

(ii)

B1

B1

Solving equation

$2 \times \text{(ii)} - \text{(iii)} \quad 14g + c = 13$

subtracting (i) $16g = 32$

$g = 2$

using (i) and (ii)

$c = 3$

$f = -3$

A1

A1

Hence required equations is

$x^2 + y^2 + 4x - 6y + 3 = 0$

$(x+2)^2 + (y-3)^2 - 10 = 0$

$(x+2)^2 + (y-3)^2 = 10$

A1 [6]

3 When $n = 1$

$$\text{L.H.S} = \sum_{r=1}^n r^2 = 1^2 = 1$$

$$\text{RHS} \frac{1}{6}n(n+1)(2n+1) = \frac{1}{6}(1)(2)(3) = 1$$

M1

$\therefore$ Statement is true for $n = 1$

A1

Assume statement is true for $n = k$ i.e. $\sum_{r=1}^k r^2 = \frac{1}{6}k(k+1)(2k+1)$

B1

When $n = k + 1$

$$\sum_{r=1}^{k+1} r^2 = \sum_{r=1}^k r^2 + (k+1)^2 \quad \text{adding the } k+1^{\text{th}} \text{ term} \quad \text{M1}$$

$$= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$$

$$= \frac{1}{6}(k+1)[k(2k+1) + 6(k+1)]$$

simplification M1

$$= \frac{1}{6}(k+1)(2k^2 + 7k + 6)$$

$$= \frac{1}{6}(k+1)(k+2)(2k+3)$$

$$= \frac{1}{6}(k+1)[(k+1)+1][2(k+1)+1]$$

A1

Since when $n = 1$ is true, when $n = k$ is true

$\Rightarrow$ It is true when $n = k + 1$. Hence the statement is true for all Z^+ by the principle of mathematical induction.

A1

[7]

4

(a) (i)

$$\frac{dy}{dx} = e^x - 8e^{-2x}$$

B1

Equating to zero

$$e^x - 8e^{-2x} = 0$$

M1

$$e^{3x} = 8$$

$$(e^x)^3 = 2^3$$

$$e^x = 2$$

$$x = \ln 2$$

$$\text{or } \frac{\ln 8}{3}$$

A1

$$y = e^{\ln 2} + 4e^{-2\ln 2}$$

$$= 2 + 1$$

$$= 3$$

 $\therefore$ coordinates are $(\ln 2; 3)$

A1

determining the nature

$$(ii) \quad \frac{dy}{dx} = e^x - 8e^{-2x}$$

$$\frac{d^2y}{dx^2} = e^x + 16e^{-2x}$$

for
second
derivative.

M1

at turning point

$$x = \ln 2$$

$$\frac{d^2y}{dx^2} = 2 + \frac{16}{4} = 6 > 0$$

 $\therefore$ stationary point is minimum

A1

$$(b) \quad 2x - 3x \frac{dy}{dx} - 3y + 2y \frac{dy}{dx} = 0$$

M1

$$\frac{dy}{dx} = \frac{3y - 2x}{2y - 3x}$$

A1

$$\text{At point } (1; 4) \quad \frac{dy}{dx} = 2$$

A1

$$\therefore \text{gradient of normal} = -\frac{1}{2}$$

$$\text{Hence equation of normal is } y - 4 = -\frac{1}{2}(x - 1)$$

$$\text{i.e. } x + 2y = 9$$

$$(c) \quad x \frac{d\theta}{dx} = \cos^2 \theta$$

$$\int \frac{1}{\cos^2 \theta} d\theta = \int \frac{1}{x} dx$$

$$\int \sec^2 \theta = \int \frac{1}{x} dx$$

$$\tan \theta = \ln x + c$$

separating and attempting to integrate

M1
A1

M1

A1 or
equivalent
[13]

$$\begin{aligned} 5 \quad (a) \quad (i) \quad \det M &= \begin{vmatrix} 1 & -1 \\ -2 & 3 \end{vmatrix} - \begin{vmatrix} 1 & -1 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix} \\ &= -2 \end{aligned}$$

M1

A1

$$\text{Adj of the matrix} = \begin{pmatrix} 1 & -5 & -2 \\ -2 & 4 & 2 \\ -1 & 1 & 0 \end{pmatrix}$$

M1 transpose

cofactors M1 correct
method
~~A2~~ correct
entries

$$\therefore M^{-1} = -\frac{1}{2} \begin{pmatrix} 1 & -5 & -2 \\ -2 & 4 & 2 \\ -1 & 1 & 0 \end{pmatrix} \text{ or}$$

$$\begin{pmatrix} -1 & 5 & 2 \\ -4 & -2 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

A1
-1

$$(ii) \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 1 & -5 & -2 \\ -2 & 4 & 2 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}$$

M1

M1

$$= \begin{pmatrix} 9 \\ -9 \\ -2 \end{pmatrix}$$

A1 All correct

$$(iii) \quad N = -\frac{1}{2} \begin{pmatrix} 1 & -5 & -2 \\ -2 & 4 & 2 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}$$

M1

$$= -\frac{1}{2} \begin{pmatrix} -9 & -1 & 0 \\ 6 & 0 & -2 \\ 3 & -1 & 0 \end{pmatrix} \text{ or } \begin{pmatrix} \frac{9}{2} & \frac{1}{2} & 0 \\ -3 & 0 & 1 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} \quad \begin{matrix} M1 \\ A1 \end{matrix} \quad [13]$$

- 6 (a) (i) If A lies on the plane then the coordinates of must satisfy the equation of the plane.

$$\text{i.e. } r \cdot \begin{pmatrix} 2 \\ -4 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -4 \\ -1 \end{pmatrix}$$

M1

$$= 6 + 4 - 2$$

$$= 8 \text{ shown}$$

$$(ii) \quad \overline{BA} = (3i - j + 2k) - (7i - 9j)$$

$$= -4i + 8j + 2k$$

$$= -2(2i - 4j - k)$$

i.e. BA is || to the normal vector

M1 - index of proper
A1 conclusion

$$(b) \quad \overline{BO} = -7i + 9j \quad \overline{BA} = -4i + 8j + 2k$$

$$\cos \hat{OBA} = \frac{(-7i + 9j) \cdot (-4i + 8j + 2k)}{\sqrt{49 + 81} \sqrt{16 + 64 + 4}}$$

M1 M1

$$= 0.9569$$

$$\therefore \hat{OBA} = 16.9^\circ \text{ or } 17$$

A1

$$0.294^c \text{ rad.}$$

$$\text{or } 163.1$$

CHAKAIPA
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$$(c) \quad \sin \theta = \frac{b \cdot \hat{n}}{(b)}$$

$$= \frac{\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -4 \\ -1 \end{pmatrix}}{\sqrt{2^2 + 2^2 + (-1)^2} \sqrt{2^2 + 4^2 + 1^2}} \quad \text{M1}$$

$$= \frac{6}{\sqrt{14} \sqrt{21}} \quad \frac{6}{3\sqrt{21}} \text{ or}$$

$$= 0.3499$$

$$= 20.25^\circ$$

$$(d) \quad \frac{\begin{pmatrix} 7 \\ -9 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -4 \\ -1 \end{pmatrix} - 8}{\left| \frac{a \cdot \hat{n} - d}{n} \right|} = \frac{(7i - 9j) \cdot (2i - 4j - k) - 8}{\sqrt{14} \sqrt{17}} \quad \text{M1}$$

$$= \frac{14 + 36 - 8}{\sqrt{17}} \quad \frac{42}{\sqrt{21}} = \frac{2\sqrt{21}}{\sqrt{21}} \quad \text{or } 2\sqrt{21} \quad \text{A1}$$

$$= 10.2 \quad \text{or } \sqrt{84} \quad \text{or } 9.17 \text{ to } 3 \text{ s.f.} \quad \text{M1}$$

$$7 \quad (a) \quad W_1 - W_2 = (1 + ix) - (x + iy) = 3i$$

$$1 - x = 0$$

$$\Rightarrow x = 1$$

$$x - y = 0$$

$$-1 - y = 3$$

$$\Rightarrow y = -2$$

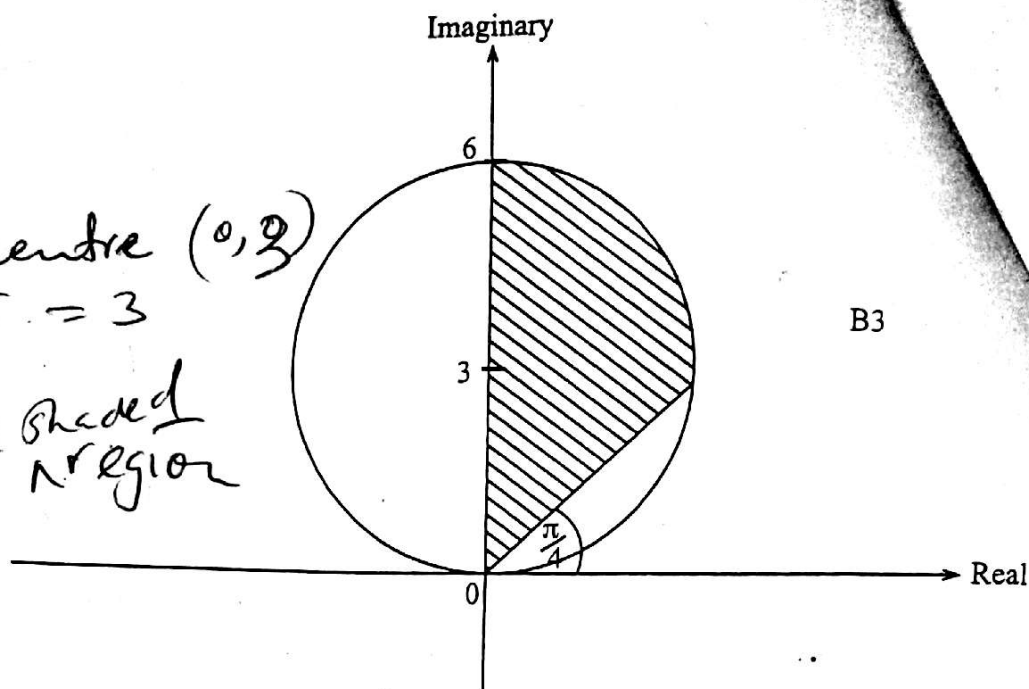
M1 equating
real
part.

A1 or
equation
in y part

A1

(b)

B₁ circle centre (0, 3)
 r = 3
 B₁ angles
 B₁ correct shaded region



B3

(c) (i) $\left(\cos \frac{1}{4}\pi + i\sin \frac{1}{4}\pi\right)^{12}$

$= \left(\cos \frac{1}{4}\pi \times 12 + i\sin \frac{1}{4}\pi \times 12\right)$ De - Moivre M1

$= \cos 3\pi + i\sin 3\pi$

$= -1$ A1

(ii) $\cos 4\theta + i\sin 4\theta = (\cos \theta + i\sin \theta)^4$ B1

$= (\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta) + i(4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta)$ M1, A1

$= \cos^4 \theta + 4\cos^3 \theta(i\sin \theta) + 6\cos^2 \theta(i\sin \theta)^2 + 4\cos \theta(i\sin \theta)^3 + (i\sin \theta)^4$

Equating real and imaginary parts

$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$ M1

$\sin 4\theta = 4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta$

$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta}{\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta}$ M1

Dividing Numerator and denominator by $\cos^4 \theta$

$$\tan \theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan \theta + \tan^4 \theta}$$

A1 (AG)

[14]

8 (i) $45 = \frac{1}{2} \times 9.81 t^2$

$$t^2 = \frac{90}{9.81} \Rightarrow t = 3.03$$

or $\sqrt{\frac{1000}{109}} \approx \sqrt{\frac{90}{9}}$ M1

A1

(ii) Horizontal vel = $\frac{15}{3.03} = 4.95$ B1

Vert vel

$$= 9.81 \times 3.03 = 29.72$$

$$29.71363323$$

$$\frac{25.71363323}{\sqrt{909}}$$

B1

Speed at Q

$$= \sqrt{4.95^2 + 29.72^2}$$

M1

$$= 30.13 \text{ (ms}^{-1}\text{)}$$

A1

$$\frac{15}{\sqrt{\frac{90}{9}}}$$

9 (a) $F = \frac{m \times a}{2.5 \times 4} = 10 \text{ N}$ B1

(b) (i) $4.5 \times 9.81 = \frac{1}{2} g t^2$ M1

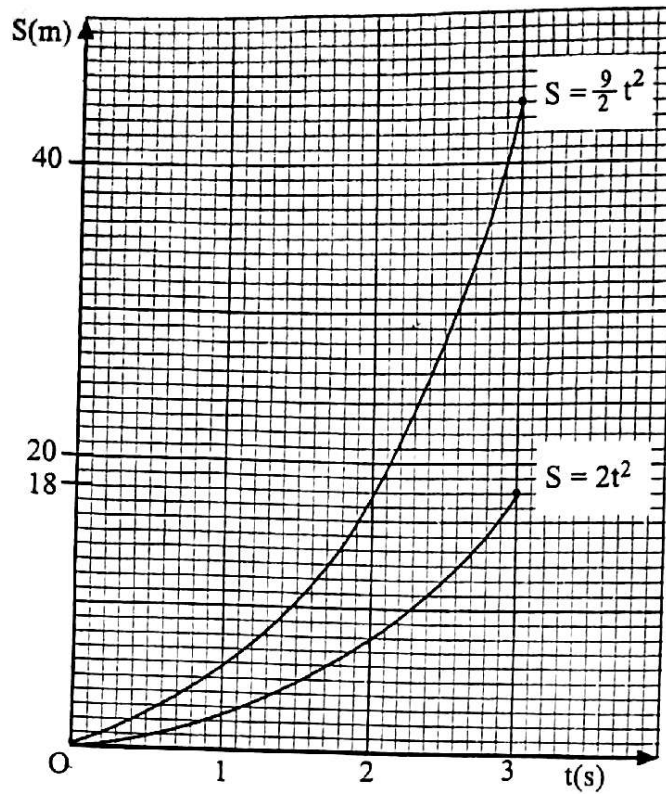
$$t^2 = 9 \Rightarrow t = 3$$

A1

(ii) $S = 0 + \frac{1}{2} \times 4 \times 9 = 18 \text{ m}$

M1A1

(c)



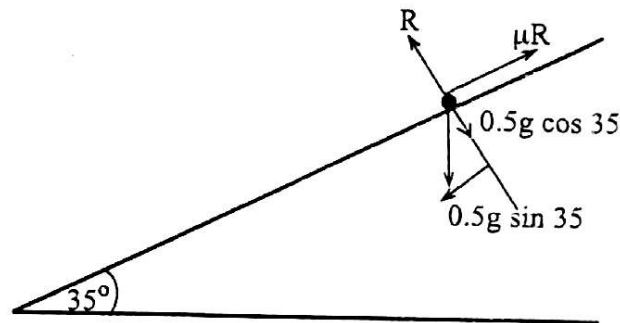
Shape B1
Shape B1
Labelling B1

~~(0,0)~~ (3, 18)

(3, 44, 1)

10

(a)



$$\perp \text{ to plane } R = 0.5 \times 9.81 \cos 35 = 4.017914$$

B1

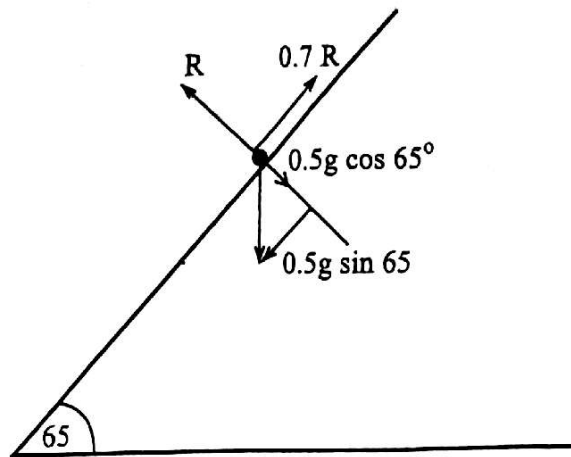
$$\parallel \text{ to plane } \mu \times 4.01794 = 0.5 \times 9.81 \sin 35^\circ$$

M1

$$\mu = 0.700$$

A1

(b)



(i) $\perp$ to plane: $R = 0.5 \times 9.81 \cos 65 = 2.07294$

B1

$\parallel$ to plane $0.5g \sin 65 - 0.7 \times 2.07294 = 0.5a$

M1

$= a = 5.99 \text{ (ms}^{-2}\text{)}$

A1

$0.8 = \frac{1}{2} \times 5.99 t^2$

M1

$t = 0.517$

A1

$v = 5.99 \times 0.517$

M1

$v = 3.097$

A1

statistics.

11 (a) $P(X \leq 7) = 1 - P(X > 7)$

$= 1 - 0.6^7$

$= 0.972$

M1

or
B₂

A1

(b) $P(X > 8 | X > 3) = \frac{P(X > 8 \cap X > 3)}{P(X > 3)}$

M1

$= \frac{0.6^8}{0.6^3}$

$= 0.6^5$

$= 0.07776$

B₃ or

M₁

A1 [5]